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HBV AND HIV COINFECTIONS AMONG INTENDING BLOOD DONORS IN PORT HARCOURT, NIGERIA

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□ *HBV is a major public health concern as it afflicts an estimated 350 million people worldwide. Studies are crucial and necessary to give us a better understanding of the epidemiology of the diseases in developing countries. A clearer picture of HBV/HIV prevalence in Africa is important in order to better educate the population and control these epidemics. This study estimated the seroprevalence of HBV/HIV coinfections among intending blood donors in Port Harcourt, Nigeria. In this cohort study, we collected blood samples from 178 intending blood donors who were confirmed to be HIV-seropositive at the University of Port Harcourt Teaching Hospital (UPTH) from September 2012 to June 2013. Commercial ELISA was used to assay for the presence of HBsAg among these subjects. The overall prevalence was found to be 6.7%. Sex and education ($P < 0.05$) were the main correlates in this study. Age was not statistically correlated ($P > 0.05$) in this study. We also observed a high overall HBV/HIV co-infection seronegativity of 93.3% among these blood donors. Group-specific seronegativity was also high ranging from 86.4–100.0%. Although the age groups (13–20, 21–35, and 36–66 years) insignificantly differed, none of their variables showed statistical association with the seronegativity. Our findings underscore the importance of screening for HBV/HIV among blood donors in developing countries, and particularly in sub-Saharan Africa, where the epidemics are still growing and a major public health concern.*

Keywords AIDS, blood donors, HIV, HBV, cohort study, prevalence, Nigeria

INTRODUCTION

Viral hepatitis is a major public health concern as hepatitis B virus (HBV) afflicts an estimated 350 million people, and hepatitis C virus (HCV) affects 150 million people worldwide.[1–4] Both viruses are endemic in sub-Saharan Africa where an estimated 75 million people (over 35 million in Nigeria) live with hepatitis B and/or C viruses.[3–6]

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Hepatitis B virus infection is associated with significant morbidity and mortality in patients with HIV infection.[4,7,8] Co-infection of HIV with HBV affects change number of patients worldwide.[4,9] Co-infection with HBV increases the risk for hepatotoxicity of HAART and likelihood of onset of an AIDS-defining illness, compared with infection with HIV-1 alone.[4,10–12] Although the HIV co-infection with HBV has been recognized worldwide in individuals exposed to blood-borne diseases, limited data are available on the extent of co-infection and effect of these viruses on the immune system in developing countries.[4,12]

In Nigeria, a country where HBV and HIV prevalence is high, HBV co-infection occurs in 10% to 70% of HIV-infected individuals.[4,13–17] A clearer picture of HIV and HBV prevalence in Africa is important in order to better educate the population, and control these epidemics.[4] Studies are crucial and necessary to give us a better understanding of the epidemiology of the diseases in developing countries like Nigeria.[3,4] In countries where HBV is highly endemic (hepatitis B surface antigen (HBsAg) prevalence rate of 8% or higher), most infections occur during infancy and early childhood.[4,18]

Therefore, this study was conducted to determine the seroprevalence of HBV/HIV coinfections among intending blood donors in Port Harcourt, Rivers State, Nigeria.

METHODS

Study Area

This study was conducted among intending blood donors attending the University of Port Harcourt Teaching Hospital (UPTH) in Alakahia, a town in Port Harcourt, the capital city of Rivers State, South south region of Nigeria between September 2012 and June 2013.

Study Population

A total of 178 intending blood donors who were confirmed HIV seropositive at the Hematology and Blood Transfusion Laboratory of the University of Port Harcourt Teaching Hospital, Port Harcourt, Nigeria were enrolled in this study. The demographic details relevant to the study were obtained as shown in Table 1.

Sample Collection

One hundred and seventy-eight blood samples were collected for this study. Venous blood was obtained into non-anticoagulated tubes. The

TABLE 1 Socio-demographic data and seroprevalence outcomes of intending blood donors tested for HBV/HIV co-infection

	Total		HBV/HIV co-infection status				
Variables	No.	%	Positives	%	Negatives	%	Statistics
Age (years)							
13-20	50	28.1	4	8.0	46	92.0	
21-35	78	43.8	6	7.6	72	92.4	
36-66	50	28.1	2	4.0	48	96.0	$P > 0.05$
Sex							
Males	78	43.8	4	5.1	74	94.9	
Females	100	56.2	8	8.0	92	92.0	$P < 0.05$
Education							
Primary	20	11.2	0	0.0	20	100.0	
Secondary	50	28.1	2	4.0	48	96.0	
Tertiary	64	35.9	4	6.3	60	93.7	
None	44	24.7	6	13.6	38	86.4	$P < 0.05$
Total	**178**	**100.0**	**12**	**6.7**	**166**	**93.3**	

samples were centrifuged at 2000 resolution per minutes (rpm) for 5 min to obtain sera. The sera were stored at −20°C for serologic assay of HIV and HBV.

Serologic Assay

Serum from each samples were assayed for HBsAg using a commercially available ELISA kit (DIA.PRO Diagnostic Bioprobes Srl Via G. Carducci n° 27 20099 Sesto San Giovanni (Milano) – Italy) according to the manufacturer's specifications. Results were interpreted according to the manufacturer's specifications.

Data Analysis

The data were recorded in and analyzed using Microsoft Excel spreadsheet (Microsoft Corporation). The seroprevalence was calculated as the number of serologically positive samples divided by the total number of samples tested. The Chi-square test was used to determine associations between seropositivity and socio-demographic variables. The level of statistical significance was set at $P \leq 0.05$.

RESULTS

One hundred and seventy-eight HIV-seropositive intending blood donors were examined in this study (Table 1). Of the 178 blood donors

tested, 12 (6.7%) were seropositive and 166 (93.3%) were seronegative for HBV/HIV co-infections. Table 1 shows the seroprevalence of HBV/HIV amongst HIV-seropositive intending blood donors in relation to their socio-demographic variables.

The sex-specific prevalence showed that females had higher prevalence of HBV/HIV (8.0%) than their male counterparts (5.1%). The study showed significant difference ($P < 0.05$) between sex and HBV/HIV coinfections amongst subjects. The age-specific prevalence showed that HBV/HIV coinfections was highest in ages 13–20 years (8.0%), followed by ages 21–35 years (7.6%) while 36–66 years had the least (4.0%), however this differences were not statistically associated ($P > 0.05$). The education-specific prevalence showed that HBV/HIV coinfections was highest in subjects with no educational background (13.6%), followed by those with tertiary education (6.3%) and secondary education (4.0%). Zero seropositivity was observed in subjects with primary level of education (0.0%). This was statistically associated ($P < 0.05$) with seropositivity of HBV/HIV coinfections amongst subjects.

We also observed a high overall HBV/HIV coinfection seronegativity of 93.3% among these blood donors. Group-specific seronegativity was also high ranging from 86.4–100.0%. Although the age-groups (13–20, 21–35, and 36–66 years) insignificantly differed, none of their variables showed statistical association with the seronegativity.

DISCUSSION

Hepatitis B virus (HBV) co-infection with HIV is associated with increased morbidity and mortality.[19,20] There is a heavy burden of HIV-HBV and HIV- HCV co infections in many regions of the developing world,[21] Nigeria inclusive.[12,22] Hepatitis co-infection with HIV accelerates disease progression in both HCV and HBV and also increases the risk of antiretroviral drug associated hepatotoxicity.[23]

In Nigeria, studies on the prevalence of HBsAg among HIV-positive patients have been documented. One hundred and seventy-eight HIV-seropositive intending blood donors were examined in this study. Of the 178 blood donors tested, 12 (6.7%) were seropositive for HBV/HIV co-infections. It is noteworthy to state that 6.7% of the subjects; ages 13–66 years tested positive to HBV/HIV co-infections. This suggests that they may have contracted the virus from their mother, family members or peer groups.[4] Also, a history of contact with jaundiced person has been identified independent risk factor for HBsAg seropositive status.[4,24]

The prevalence of HBV/HIV co-infections among blood donors in Port Harcourt, Nigeria is 6.7%. This 6.7% reported in this study is far lower than the 30.6% value reported by Ojo et al.[4] It is also lower than the 25.0% reported by Uneke et al.[25] among HIV-infected patients in Jos, Nigeria.

It is lower than the 28.4% reported among HIV positive donors in Ikeja, Nigeria.[19] The observed HIV/HBV coinfection seropositivity (6.7%) in this study is not comparable with previous reported high figures in different parts of Nigeria and outside. Some previous reports of HIV/HBV coinfection of 15.0% in Maidugri,[26] 20.6% in Keffi.[12] 28.7% in Jos,[27] 30.4% in Ilorin,[28] 70.5% in Kano,[15] 33.8% in India,[29] 9.2% in Lagos,[30] 9.7% in Niger Delta,[13] and 8.7% in Thailand[31] are, however, lower than observed figures in this study. This low prevalence is lower compared to 19.8% reported by Italian researchers.[32] Similarly, the result in this study is very low compared to 25.5% reported by French researchers.[33] Varying sample size, test kit sensitivity, and specificity may be responsible for the differences in prevalence figures in this group of patients.[19]

This 6.7% reported in this study is comparable with the 5.4% prevalence reported by Opaleye et al.[34] among blood donors in Benin City, Nigeria. It is also comparable to the 6.0% reported among HIV negative blood donor controls in Ikeja, Lagos, Nigeria.[19] The differences in prevalence in these studies could be attributed to differences in patient selection.[4] HBV infection rate in this study is comparable to what is seen in the HIV seronegative population, which indicates an endemic infection by the hepatitis B virus in the Nigerian population.[3,4,34] The co-infection rate of HBV/HIV is low (6.7%) in this study, and is compared to what is found by other investigators in Nigeria.[2–4,34,35] The 6.7% co-infections rate observed in this study is also comparable to the 7.9% co-infection rate of HBV/HIV reported by Tremeau-Bravard et al.;[3] the 6.0% reported from South Africa;[36] and the 6.0% reported among the patients attending Aminu Kano Teaching hospital (AKTH).[37] This is low prevalence compared favourably to the 6.3% reported by Abdel-kader et al.[38] in HIV infected patient in Australia. Our value also compared favourably to the 6.9% (6.6–7.2%) reported by Price et al.[20] in UK CHIC. However, the 6.7% than the 3.9% dual presence of HBsAg and anti-HCV reported in HIV infected patients by Balogun et al.[19] in Ikeja, Nigeria.

The sex-specific prevalence showed significant difference ($P < 0.05$) between sex and HBV/HIV coinfections among subjects. The reason for this difference might be due to the larger number of females (53.2%) in this study. Other studies in Nigeria also reported that more females than males visit hospitals for medical attention.[25] This may be due to the fact that females at younger age group are most sexually active. This statistically associated sex-specific prevalence suggests that both sexes were not equally susceptible to HBV/HIV co-infections. It also points to the fact that gender might necessarily be an important epidemiological determinant of HBV/HIV co-infections among blood donors. Balogun et al.[19] reported that the prevalence of hepatitis co infection is higher among the males than the female blood donors in Ikeja, Lagos, Nigeria.

The statistical association of sex in HBV/HIV co-infections in the present study suggests that they were not equally exposed to HBV/HIV. This corroborated with earlier findings in Nigeria,[24,39] however, it contradicts with what was reported by other authors elsewhere.[20–43] Our finding also disagrees with what was reported by Mehmet et al.[44] that male sex was an important risk factor for HBsAg positivity. Mehmet et al.[44] reported significantly higher prevalence in males than females in both rural and urban areas. Our study also disagrees with previous reports in Northern Nigeria [15] and elsewhere.[45–47] Taura et al.[37] also reported highest percentage prevalence in females as compared with males.

The age of acquiring infection is the major determinant of the incidence and prevalence rates[4,48] and serological evidence of previous HBV infections varies depending on age.[4,24] However, age was not statistically correlated in this study. HBV/HIV coinfections was higher in ages 13–20 years (8.0%) and 21–35 years (7.6%) and least in ages 36–66 years (4.0%). This collaborates with Motta-Castro et al.[49] who also reported that age was not significantly associated with HBsAg seropositivity among Afrodescendant community in Brazil. In disagreement with our finding, Taura et al.[37] reported age associated HBV/HIV coinfection with highest percentage prevalence in age group 21–30 years of age. Price et al.[20] reported that the factors associated with incident HBV/HIV coinfection were older age and IDU.

Also, serological evidence of previous HBV infections varies depending on socioeconomic class.[4,24] In this study, educational level was statistically correlated as HBV/HIV coinfections was higher in subjects with no educational background (13.6%) than their counterparts with tertiary education (6.3%) and secondary education (4.0%). Zero seropositivity was observed in subjects with primary level of education (0.0%).

In this study, we also observed a high overall HBV/HIV coinfection seronegativity of 93.3% among these blood donors. Group-specific seronegativity was also high ranging from 86.4–100.0%. Although the age groups (13–20, 21–35, and 36–66 years) insignificantly differed, none of their variables showed statistical association with the seronegativity. However, asymptomatic HBV infection among HIV seropositives without proper identifiable risk factors or mode of acquisition calls for general surveillance, mass immunization, and public health education to curtail the spread of the virus and its sequalae.[24]

CONCLUSION

The prevalence of HBV in this HIV seropositive blood donors is in line with estimates from other studies and is low by international standards. This study further confirms the presence of HBV/HIV co-infections (6.7%)

among intending blood donors. The study has also shown higher prevalence of co-infections of viral hepatitis with HIV in females than their male counterparts. Age was not statistically correlated in this study. Sex and education were the main correlates in this study. The relatively high co-infection rate of HBV/HIV among younger intending blood donors in this study indicates that most of them may have acquired the infection through sex. Possibility also exists of an ongoing horizontal spread of the infection. This calls for public health education to enlighten the blood donors and general public in Port Harcourt of the possible risk factors and routes of infection.

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